

National Park Service

JEAN LAFITTE NATIONAL HISTORICAL PARK & PRESERVE



Statement of Work

Chalmette National Cemetery Perimeter Wall Vegetation
Removal, and Cleaning
326695A

1.00 SCOPE

To clean and remove vegetation from perimeter brick masonry walls in preparation for a brick masonry repair campaign.

The cemetery is bound by three sections of brick wall at its east, west and south perimeters. The existing brick wall is four wythes thick at its base, has a common bond inset panel of two wythes, and a two wythe overhang in a stretcher arrangement. Rowlock brick coping lies at the top. Each panel section is separated by piers with pyramid caps with a brick pilaster at mid span. The brick wall has been subject to multiple restoration and reconstruction efforts, the last two having taken place in the 2010's and 1980's. A typical wall plan and elevation are shown on the following page.

2.00 REQUIREMENTS

1. The Contractor shall furnish all supervision, labor, materials and equipment necessary to accomplish the work of this contract.
2. The contractor is responsible for field-verification of any drawings that are provided by the government that indicate quantitative or dimensional data.
3. The contractor shall follow all of the Park's policies, procedures and guidance with regard to performing work activities on federal lands.
4. Location of Work:
Chalmette National Cemetery
8606 W St Bernard Hwy
Chalmette, LA 70043
5. All work conducted in association with this contract shall meet or exceed industry standards with regards to the use and selection of materials and the methods utilized to accomplish each task.
6. Masonry Cleaning Trades Qualifications:
 - a. The Masonry Cleaning Trade must have demonstrated proficiency in the restoration of historic brick masonry by the successful performance of work of the nature specified on at least three historic sites in the last ten years, the foregoing installations having been the complete and undivided responsibility of the Masonry Cleaning Trade, and the work of the foreman proposed for this project. Submit the following:
 - i. Name of Foreman and Certificate of Training.
 - ii. Name, location and description of the three representative projects.
 - iii. Name and phone number of Owner, Architect and Contractor references for each of the three representative projects.
 - b. A qualified Masonry Cleaning Trade will need to be engaged to perform the work within the attached specification.
 - c. It is required that the work of this Section be the responsibility of a single Masonry Cleaning Trade.
 - d. Only skilled journeymen masons who are familiar and experienced with the materials and methods specified and are familiar with the design requirements shall be used for masonry restoration.
 - e. One skilled journeyman mason, designated as foreman, shall be present at all times during masonry restoration and shall personally direct the work.

4.00 WORK TO BE PERFORMED

4.01 SECTION INCLUDES

A. Cleaning existing masonry - The entire building shall be cleaned. Provide separate cost for east and west walls separately.

4.02 SUBMITTALS

A. Product Data:

1. Submit for each cleaning agent (detergent, chemical, etc.).
2. Submit MSDS for each cleaning agent.
3. Submit manufacturer's detailed application instructions for proprietary cleaners.

B. Submit a description of proposed protection of surrounding materials and site features including grave markers and landscape. Describe control of runoff during operations. Describe in detail the materials, methods, and equipment to be used.

C. If materials and methods other than those indicated are proposed for cleaning work, provide a written description, including evidence of successful use on other comparable projects, and a testing program to demonstrate their effectiveness for this Project.

D. Trade contractor qualifications.

1. Trade contractor must have demonstrated proficiency in the restoration of historic properties by the successful performance of work of the nature specified on at least three historic properties in the last ten years, the foregoing installations having been the complete and undivided responsibility of the trade contractor and the work of the foreman proposed for this project. Submit the following:
 - a. Name of Foreman and Certificate of Training.
 - b. Name, location and description of the three representative projects.
 - c. Name and phone number of Owner, Architect and Contractor references for each of the three representative projects.

4.03 QUALITY ASSURANCE

A. Mockups: The mockup process shall be used to determine the cleaning products and methods to be used. The gentlest means possible to clean each condition shall be used. The mockup shall first test the effectiveness of the specified Liquid Detergent, Site-Mixed Detergent-Fungicide and Proprietary Biodegradable, pH neutral, Liquid Cleaning Solution. If these products are not determined to be effective, after review by the Architect, the mockup shall test the effectiveness of the specified Proprietary Liquid Cleaners.

1. Prepare field samples for cleaning procedures to demonstrate aesthetic effects and qualities of materials and execution. Use materials and methods proposed for completed Work and prepare samples under same weather conditions to be expected during remainder of Work.
2. Locate mockups on the building where directed by Architect.
3. Prepare a series of samples representing each of the following surface conditions:
 - a. Cleaning of masonry surface.
 - b. Cleaning of porch tile floor.
4. Mock-up of cleaning to be 3'x3'. Mock-up of coating removal shall consist of one entire element of each type indicated. Mock-up may require multiple applications to achieve acceptable results.
5. After review by the Architect, prepare additional samples, adjusting materials, methods, timing, etc., as necessary to clean masonry to required finish. Perform mock-up using each cleaning agent specified for each substrate.

- a. Test cleaners and methods on samples of adjacent materials for possible adverse reactions, unless cleaners and methods are known to have a deleterious effect.
 - b. Allow a waiting period of not less than 14 days after completion of sample cleaning to permit a study of sample panels for negative reactions.
- 6. Schedule mock-ups sufficiently in advance of scheduled work to allow for multiple adjustments of mock-up without impacting construction schedule.
- 7. Obtain Architect's approval of mockups before starting the remainder of restoration and cleaning.
- 8. Maintain mockups during construction in an undisturbed condition as a standard for judging the completed Work.
- B. It is required that the work of this Section be the responsibility of a single Trade Contractor.
- C. Trade contractor: Work must be performed by a firm having not less than 5 years successful experience in comparable masonry restoration projects and employing personnel skilled in the restoration process and operations indicated.

4.04 DEFINITIONS

- A. All pressures measured at discharge end.
- B. Garden Spray: Spray of hand-pump-up garden-type ("Hudson") sprayer with nozzle adjusted to a cone-shape. Powered garden-type sprayers providing equivalent spray are also acceptable. Stainless steel or plastic parts required (galvanized not acceptable).
- C. Very-Low Pressure Spray: 30 psi (nominal) through a 3/4-inch diameter hose fitted with a nozzle producing a conical spray of approximately 60 degrees applied at a distance not closer than 4 feet from the surface. Provide pressure/volume/cut-off valve at discharge end.
- D. Low-Pressure Spray: 100 to 200 psi; 4 to 6 gpm.
- E. Medium-Pressure Spray: 200 to 600 psi; 4 to 6 gpm.
- F. High-Pressure Spray: 600 to 1200 psi; 4 to 6 gpm.

4.05 DELIVERY, STORAGE, AND HANDLING

- A. Deliver other materials to Project site in manufacturer's original and unopened containers, labeled with type and name of product and manufacturer.
- B. Comply with manufacturer's written instructions for minimum and maximum temperature requirements for storage.

4.06 PROJECT CONDITIONS

- A. Clean surfaces only when air temperature is 40 degrees F and above and will remain so for at least 7 days after completion of cleaning.

4.07 SEQUENCING AND SCHEDULING

- A. Perform masonry cleaning and restoration work in the following sequence:
 - 1. Remove vegetation adjacent to and within cemetery walls..
 - 2. Install temporary materials where required to prevent entry of water or chemicals into interior of masonry work..
 - 3. Clean existing soil, stains, efflorescence, etc., from masonry.

5.00 PRODUCTS

5.01 MATERIALS

- A. Water: Clean, potable water.
- B. Warm water for mixing cleaning solutions.

- C. pH paper with 3 colors to identify numeric pH level.
- D. Liquid Detergent: Liquid dish washing detergent such as "Ivory Clear".
 - 1. Shall not contain additives (whiteners, brighteners, detergents, fragrance, etc.).
- F. Powdered Detergent: Powdered detergent such as "Tide".
 - 1. Shall not contain additives (whiteners, brighteners, detergents, fragrance, etc.).
- E. Trisodium Phosphate:
 - 1. Strong, base-type, phosphate-containing, powdered cleaning material.
 - 2. Other chemical or common names include Sodium Orthophosphate; Tribasic sodium phosphate; Trisodium orthophosphate; TSP; Phosphate of soda.
 - 3. Available from chemical supply house, grocery store, supermarket, or hardware store.
 - 4. Shall not contain additives (whiteners, brighteners, detergents, fragrance, etc.).
- D. Sodium Hypochlorite (NaOCL).
 - 1. Other chemical or common names include bleaching solution; household bleach; laundry bleach; solution of chlorinated soda.
 - 2. Available from chemical supply house, grocery store, supermarket, hardware store, or janitorial supply distributor.
 - 3. Shall not contain additives (whiteners, brighteners, detergents, fragrance, etc.).
- G. BASIS OF DESIGN: Proprietary Biodegradable, pH neutral, Liquid Cleaning Solution:
 - 1. D/2 Biological Solution, www.d2bio.com or (917) 693-7441.
- H. Proprietary Liquid Cleaners:
 - 1. Enviro Klean ReVive; Prosoco, Incorporated.
 - 2. MasonRE B All Purpose Cleaner; Cathedral Stone.
 - 3. Envirestore 100TM; Dietrich Technologies.

5.02 MIXES

- A. Liquid Detergent:
 - 1. Mix:
 - a. 1/2 fluid ounce liquid detergent.
 - b. 1 gallon warm water.
- B. Site-Mixed Detergent-Fungicide:
 - 1. Mix the following until completely dissolved:
 - a. 1/3 cup trisodium phosphate cleaner.
 - b. 1/3 cup powdered detergent.
 - c. 1 qt 5% sodium hypochlorite bleach.
 - d. 3 qts. warm water.
- C. Proprietary Biodegradable, pH neutral, Liquid Cleaning Solution:
 - 1. Mix in accordance with manufacturer's instructions.
- D. Proprietary Liquid Cleaners:
 - 1. Mix in accordance with manufacturer's instructions.

5.03 TEMPORARY COVERS

- A. Liquid Strippable Masking Agent: Manufacturer's standard liquid, film-forming, strippable masking material for protecting glass, metal, and polished masonry surfaces from the damaging effects of acidic and alkaline masonry cleaners.
- B. Polyethylene Sheet.
- C. Adhesive Tape: Non-staining, leaving building surface residue-free after tape is removed.

5.04 EQUIPMENT

- A. Cleaner Sprayer: Very low pressure pump sprayer such as a garden sprayer.
- B. Water Sprayer: Very low pressure sprayer (hose and nozzle) or very low pressure power

- assisted sprayer.
- C. Natural or nylon bristle brushes. Metal not permitted.
- D. Wood or plastic scrapers. Metal not permitted.
- E. Other equipment as necessary for the Work.

6.00 EXECUTION

6.01 VEGETATION REMOVAL IMMEDIATELY ADJACENT OR WITHIN THE WALL ASSEMBLY

- A. Using lopping or pruning shears, cut vine trunk(s) about four inches (4") above grade and remove a section of stem above this cut. Leave approximately a six inch (6") gap between the stump of the trunk and the stem.
- B. Make vertical cuts through the bark of the stump and peel the bark back slightly to expose about one inch (1") of the inner wood.
- C. Apply a thick paste made from ammonium sulfamate crystals, or spray apply weed killer to the exposed surfaces of the stump. **BE CAREFUL NOT TO GET ANY ON THE MASONRY.**
- D. Allow the vines to die naturally, then remove the dried tendrils and roots, taking care not to remove bits of masonry and mortar.
- E. Large, heavy, mature vines may be more easily removed by cutting the network of stems into smaller sections.
- F. If the root system of mature vines is extensive, it may be necessary to remove individual roots from each mortar joint.
- G. Do not leave any dead wood within joints of the wall. As the wood decays, voids may be left with the wall that can lead to more serious problems.
- H. Gently scrub the wall with a stiff, non-metallic bristle brush and clean, clear water (sprayed at low pressure) to remove any remaining dried plant material.
- I. For root pods that are more difficult to remove, try using low pressure steam while brushing with a stiff, non-metallic bristle brush.

6.02 PROTECTION

- A. Control of Runoff:
 - 1. Do not allow cleaners and rinses to collect, pond, or form soft muddy conditions at the base of the building that do not dissipate within 24 hours.
 - 2. Provide temporary gutters, diversions, collection basins, pumps, and piping if necessary to direct excess liquid to approved discharge points.
- B. Protection of Vegetation: A portion of the existing plant life is indicated elsewhere in the Contract Documents to remain. Do not allow cleaners and rinses to contact vegetation to remain.
- C. If inadvertent spills of cleaner contact vegetation or other building elements, rinse immediately with potable water until free of cleaner.
- D. Do not apply sprays during windy conditions sufficient to carry overspray into contact with other surfaces, vegetation, or people.

6.03 CLEANING STANDARD REQUIRED:

- A. General:
 - 1. Clean using products and procedures established through the mockup process.
 - 2. Clean masonry to remove mildew, fungus, lichens, algae, dirt, soot, and other forms of soiling without damage to masonry (whether visually apparent or not) and without change in appearance of clean masonry (whether color, texture, polish, finish, etc.) other than removal of the soil.
 - 3. Degree of cleanliness in the Work shall match that achieved in the approved mock-ups.

6.04 CLEANING, GENERAL

- A. Identify panels of the wall to be cleaned sequentially.
- B. Proceed within each panel from the base of the wall to the top, unless otherwise approved.
- C. Prewetting:
 - 1. As cleaning proceeds upwardly, maintain lower portions and immediately adjacent portions continuously wet and streak-free and soil-free.
 - 2. Extend the wetted area horizontally beyond the immediate area to be cleaned.
 - 3. Wet the area beneath the area to be cleaned, from grade level up to the area to be cleaned.
 - 4. Maintain these adjacent areas wet with water until rinsing is complete to avoid streaking and deposition of cleaners and residues onto adjacent surfaces.
- D. Thoroughly remove cleaners by rinsing with potable water. A final rinse shall be performed from the top of the building down to the base of the building.
- E. Clean masonry surfaces in a uniform manner. Include flat surfaces, cornices, moldings, ornament, recesses, tops and undersides, etc., to produce a uniformly clean result.
- F. Do not apply different cleaners on a given area unless the cleaner used previously has been thoroughly washed away.
- G. Adjustments to meet Project Conditions:
 - 1. Repeat cleaning procedures or adjust dwell times or adjust the amount or type of scrubbing effort or adjust concentration of cleaners (or a combination the preceding), depending upon the amount and type of soil or stain present on the various parts of the building, and so as to achieve a uniformly clean result and without change in Contract Time or Price.
 - 2. Obtain the Architect's approval of such adjustments.
 - 3. Do not exceed concentrations or dwell times or repeat procedures beyond the limits specified or approved by the Architect.

6.05 SPRAYS

- A. Do not use power-assisted spray without the written authorization of the Architect.
- B. Provide very low pressure spray, taking water from hose bibbs to portions of the building required to be cleaned.
- C. If the Architect determines that unassisted pressure at hose-end from the building water supply does not provide adequate pressure or volume, provide power-assisted spray adjusted to simulate very-low pressure spray without change in Contract Time or Price. Obtain the written authorization from the Architect.
- D. If the Contractor so requests and the Architect determines that due to remote location or configuration or other Project factors, it is impracticable to use hoses to rinse selected portions, provide power-assisted spray adjusted to simulate very-low pressure spray without change in Contract Time or Price. Obtain the written authorization from the Architect.

6.06 APPLICATION OF LIQUID CLEANERS

- A. Remove as much plant growth as possible using a knife blade and stiff bristle brush. Drybrush the surface before wetting to remove bulk growth.
- B. Pre-wet the area to be cleaned (and the adjacent areas) with a water spray.
 - 1. Extend the wetted area horizontally beyond the immediate area to be cleaned.
 - 2. Wet the area beneath the area to be cleaned, from grade level up to the area to be cleaned.
 - 3. Maintain these adjacent areas wet with water until rinsing is complete to avoid streaking and deposition of cleaners and residues onto adjacent surfaces.

C. Apply the solution to the affected area using either a garden spray or medium-stiff natural bristle brush. Use large, flat brushes for flat areas; use small brushes to access recesses, reveals, and detail of ornament.

1. Scrub with a natural or artificial bristle brush and allow to dwell as necessary depending on degree of soiling and application temperatures.
2. Dwell times are estimated to be 20 to 30 minutes between 40 and 70 degrees F, and 10 to 15 minutes at 70 degrees F and above, but may range up to an hour or longer depending upon degree of soiling, scrubbing effort, and other factors. Consult manufacturer for required dwell time for the product being used.
3. Do not allow cleaners to dry out. Reapply cleaner or mist with water to keep the surface saturated, and scrub periodically until the growth, stain, or soil is removed.

D. After-Wash:

1. Thoroughly rinse cleaner from surface with low-pressure spray water.
2. Immediately apply after-wash to surface and allow to dwell for 3 to 5 minutes.

E. Thoroughly rinse the surface with low-pressure spray water.

1. Test liquid rinse run-off drops with pH paper to ensure that cleaning solutions have been effectively removed. Continue rinsing until pH is neutral. (pH testing of liquid detergent is not required or effective.)
2. Allow to dry.
3. Test as often as necessary to ensure reliable, repeatable results and when otherwise requested by the Architect.

F. Use prepared solutions within 24 hours.

6.07 REMOVAL OF POST-CLEANING EFFLORESCENCE

A. If a light dusting of efflorescence appears after the cleaning, remove the efflorescence by dry brushing with a clean brush. If dry brushing is not completely successful, wet the surface with water and brush to remove the efflorescence.

7.00 WORKING HOURS

Available Work Hours: Monday-Friday, 7:00 a.m. - 3:00 p.m. only. A proposed service schedule shall be developed by the contractor and approved by Contracting Officer's Representative (COR) prior to any services being performed. No work shall be performed during observed Federal Holidays.

The ten (10) Federal Holidays observed are:

New Year's Day - January 1st
Martin Luther King's Birthday - 3rd Monday in January
Presidents Day - 3rd Monday in February
Memorial Day - Last Monday in May
Independence Day - July 4th
Labor Day - 1st Monday in September
Columbus Day - 2nd Monday in October
Veterans Day - November 11th
Thanksgiving Day - 4th Thursday in November
Christmas Day - December 25th

8.00 SAFETY REQUIREMENTS

1. The contractor will enforce an effective accident prevention program and provide for a safe and healthful environment for all contractor employees.
2. The contractor and their employees will comply with all Occupational Safety and Health Administration (OSHA) standards at all times.

3. The contractor is responsible for helping safeguarding the project site as work is being performed. Appropriate signs, warning devices, and barriers shall be used as appropriate to protect the public, government and contractor employees.
4. The Contractor shall ensure that all reasonable precautions are taken to protect the public in the areas where work is being performed.
5. The Contractor shall immediately report to the COR any potential hazards found by his employees on the project.
6. The Contractor shall not allow any persons, except employees, into work areas that are not otherwise open to the public; nor shall the Contractor transport in Contractor work vehicles any persons, except employees, during times the Contractor is on the project performing work under this contract.
7. The Contractor shall use Job Hazard Analyses (JHA's) as one of many tools to insure the safe performance of work. It is the Contractor's responsibility to prepare the JHA. A JHA must be prepared by the Contractor and approved by the COR for each major work task of this contract, prior to the Contractor performing the work.
8. Prior to performing work with inherent hazards, the Contractor shall conduct a safety meeting and review the approved JHA with those employees who will be performing the work. The Government may request advanced notification of the time and date of the above JHA Tool Box safety meeting for Contractor employees, so that the Government may attend such reviews. A copy of the approved JHA's shall be available at all times to workers at the job site and a copy shall be provided to the Government.

9.00 ACCIDENT REPORTING

1. All reportable accidents, which are those resulting in death, occupational disease, traumatic injury to employees or the public, fires, and property damage by accident in excess of \$100, shall be reported to the Contracting Officer and COR immediately. Within 7 days of a reportable accident, the contractor shall fill out and forward to the Contracting Officer an Accident/Property Damage Report (Form CM-22).
2. If the Government determines that the Contractor did not use reasonable care for Government property and caused damage to or loss of said property, the Contractor shall repair the damage or replace the property, at no cost to the Government. If the Contractor fails or refuses to make such repairs or replacement, the Contractor shall be liable for the cost, which will be deducted from the contract payment.

10.00 PHYSICAL SECURITY

1. The contractor shall cooperate with Government personnel in safeguarding Government facilities and property.
2. The contractor will secure all doors or gates given access to after the completion of contracted work activities.
3. The contractor shall be responsible for reporting to the COR all acts of vandalism, larceny, pilferage, or any other damage that is discovered. Reports shall be made no later than within 24-hours after discovery. Notification must be made via telephone or in person and followed-up in writing.

11.00 ENVIRONMENTAL PROGRAM

1. The Contractor shall comply with all Federal, State, and local laws, regulations and standards regarding environmental protection.
2. The Contractor shall conduct all contracted activities in a manner to prevent contamination of Government land and property.

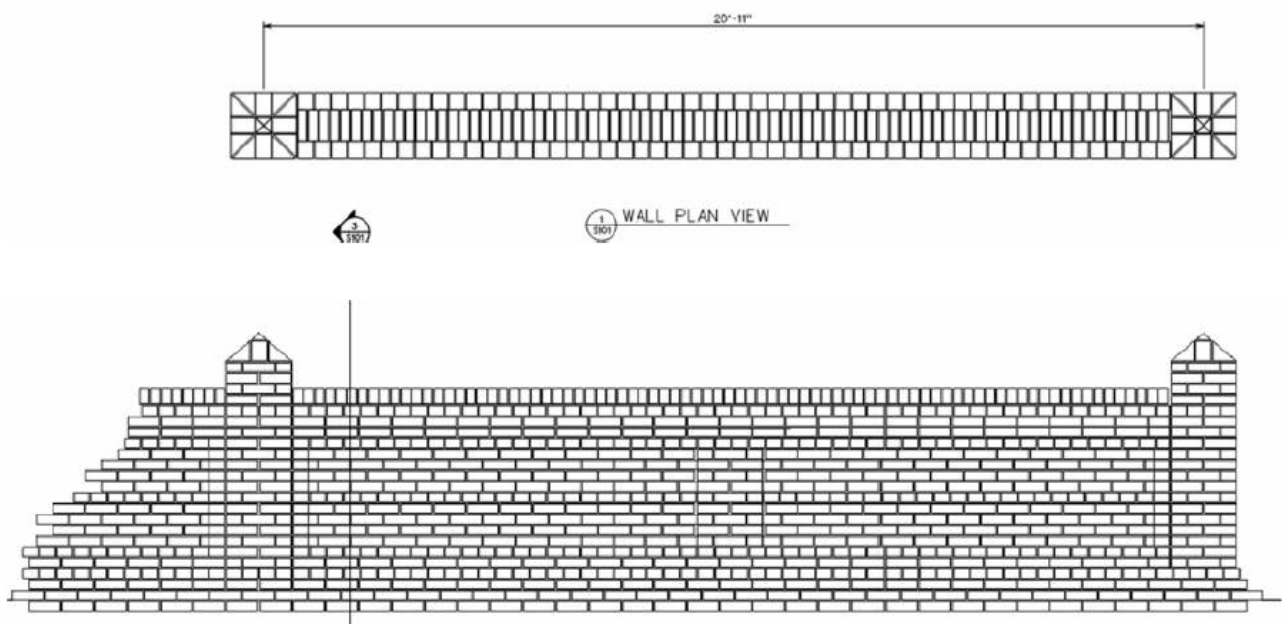
3. The Contractor shall ensure all lubricants, petroleum products, cleaning materials, or any other hazardous substances are not spilled, leaked or disposed of on Government property.
4. The contractor shall dispose of all hazardous substance in accordance with environmental regulations.
5. If spills or contamination occur, the Contractor shall be responsible for cleanup and any associated costs in accordance with Federal, State and Local Law. Such spills shall be reported to the COR immediately.
6. The Contractor shall identify all hazardous waste associated with conducting required work items, and properly label, handle, and dispose of such waste in accordance with applicable regulations.
7. The Contractor shall address hazardous materials and hazardous waste on a Job Hazard Analysis form prior to commencing work contained within this contract.

12.00 IDENTIFICATION OF CONTRACTOR EMPLOYEES AND VEHICLES

1. The contractor shall furnish their employees that are performing work under this contract with a uniform shirt and outer garments that include the company's name as identification. Contractor employees shall wear company identifying clothing at all times when performing work under this contract. Any subcontracted work shall follow the same guidelines, whereby the subcontracted company's employees shall also have company identifiable clothing worn at all times while performing work on NPS property.
2. All contractor owned vehicles that are to be used while performing work under this contract shall have the company's name prominently displayed in not less than two inch letters on the outside of both front doors of each vehicle in such a manner as to provide a readily visible means of identification.

13.00 MISC. DRAWINGS AND EXISTING CONDITIONS PICTURES (NOT ALL AREAS INCLUDED IN THESE PICTURES)

This are for reference only, contractor shall field verify conditions. Biological growth was prevalent throughout, particularly at horizontal surfaces, or at vertical faces where water may drain down. It was observed at brick and mortar, including significant discoloration of various types.



Existing Condition Photographs



Typical Wall panel



Adjacent overgrowth/



Plants growing within the wall assembly.



Significant biological growth on wall caps, typical.

14.0 CONTRACT BID SCHEDULE

Jean Lafitte National Historical Park and Preserve
Chalmette National Cemetery wall cleaning and vegetation removal.

Submit bid for all items; failure to do so may render the bid non-responsive.

ITEM NO	QUANTITY & UNIT OF MEASURE	AMOUNT OF BID
BID ITEM # 1 – Vegetation removal	L.S. Lump Sum = \$	
BID ITEM # 2 – Cleaning of east wall	L.S. Lump Sum = \$	
BID ITEM # 3 – Cleaning of west wall	L.S. Lump Sum = \$	
BID ITEM # 4 – Cleaning of south wall	L.S. Lump Sum = \$	

Award will be made to one Bidder in accordance with the Instructions to Bidders.

END